

写在前面:

本份笔记主攻应试及格(竞赛课大家都懂)。前面是控代可能出现的常见大题解法,最后补充笔记有一些判断题的易错点。

内容整体很应试,如果想选自动化建议好好学一下,看 wjd 学长的笔记之类的。

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本人最终 96.3/100, Rank 7/126 . (2026春).

(说实话,2026春卷子几乎1:1复刻了2025春,导致大家都很高分.之后不一定会有这么幸运了?)

第二章

一. 判断 V 是否为向量空间

{ 包含 $\vec{0}$
加法封闭
乘法封闭

八大公理.

二. 利用线性性质证明元素属于空间: (证明不属于空间可用反证)

eg.

3. Let X and Y be subspaces of a vector space V .

(a) Let $v \in V, v \notin X$. Prove that if $x \in X$ then $x + v \notin X$.

proof: Let $v \in V, v \notin X, x \in X$.

Assume that $x + v \in X \rightarrow$ 反证法

Since $x \in X \Rightarrow -x \in X$

$-x \in X, x + v \in X \Rightarrow x + v - x \in X$

} $\rightarrow$ 线性变换

which implies $v \in X$.

which contradicts the given premise that $v \notin X \rightarrow$ 反证法的套话.

Thus, $x + v \notin X$.

三. 证明向量组是一组基

{ 线性独立 $\rightarrow C_1 v_1 + \dots + C_n v_n = 0$ only if $C_1 = C_2 = \dots = C_n = 0$
可张成空间 $\rightarrow U$ 中任一向量可以被表示为基的线性组合.

四. 多项式空间构造基的方法.

eg. Let $U = \{p \in P_4(F), p(2) = p(5)\}$. Find a basis of U .

首先, $\dim P_4(F) = 4 + 1 = 5$. 由于每一个约束 $p(2) = p(5)$, 维数 -1 . $\dim U = 4$.

思路: 找 4 个不同阶的单项式 $\in U$ (必定线性独立)

首先 $1 \in U$. (degree = 0).

令 $p(2)=p(5)=0$: 构造一个多项式使 2,5 是根

$$\Rightarrow (x-2)(x-5) \in U. \quad (\text{degree}=2)$$

$$\text{打靶: } \begin{cases} x(x-2)(x-5) \in U & (\text{degree}=3) \\ x^2(x-2)(x-5) \in U & (\text{degree}=4) \end{cases}$$

$\Rightarrow$ 找到 4 个基.

五. 寻找直和补空间.

eg.

1. Suppose

$$U = \{(x, x, y) \in \mathbb{R}^3 : x, y \in \mathbb{R}\}$$

Find a subspace W of $\mathbb{R}^3$ such that $\mathbb{R}^3 = U \oplus W$.

$$\dim U = 2. \Rightarrow \dim W = 3 - 2 = 1$$

我们需要: ① $\dim W = \dim \mathbb{R}^3 - \dim U$

$$\textcircled{2} U \cap W = \{0\}$$

So we can choose $W = \{(x, 0, 0) \in \mathbb{R}^3 : x \in \mathbb{R}\}$
 $= \text{span}\{(1, 0, 0)\}.$

六. 向量空间寻找基

eg.

(b)

$$\left\{ \begin{bmatrix} x+3y-z \\ 4x+5y+3z \\ 3x+6z \\ -x+7y-9z \end{bmatrix} : x, y, z \in \mathbb{R} \right\}$$

(c)

$$\{(x, y, z, w) : x - 4y + 3w = 0\}$$

$$(b): \begin{bmatrix} x+3y-z \\ 4x+5y+3z \\ 3x+6z \\ -x+7y-9z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 3 \\ -1 \end{bmatrix} x + \begin{bmatrix} 3 \\ 5 \\ 0 \\ 7 \end{bmatrix} y + \begin{bmatrix} -1 \\ 3 \\ 6 \\ -9 \end{bmatrix} z$$

$$\text{let } v_1 = \begin{bmatrix} 1 \\ 4 \\ 3 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 5 \\ 0 \\ 7 \end{bmatrix}, v_3 = \begin{bmatrix} -1 \\ 3 \\ 6 \\ -9 \end{bmatrix}$$

$\rightarrow$ 但是它们不是线性相关, 所以
要做行变换.

$$[v_1, v_2, v_3] \xrightarrow{\text{row operation}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

which indicates $\begin{cases} v_1, v_2 \text{ is linearly independent} \\ v_3 = 2v_1 - v_2. \end{cases}$

So $\{v_1, v_2\}$ is a basis.

(c) $x-4y+3w=0 \rightarrow$ 通过这个可以解出一个未知数.

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 4y-3w \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \end{pmatrix} y + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} z + \begin{pmatrix} -3 \\ 0 \\ 0 \\ 1 \end{pmatrix} w$$

显然线性无关. 所以这三个就是基.

七. 过渡矩阵计算, 换基.

eg. [已知新、旧基换算关系]

4. Consider two ordered bases $E = \{v_1, v_2, v_3\}$ and $F = \{w_1, w_2, w_3\}$ for a vector space V , and suppose

新、旧基换算关系:

$$v_1 = -w_1 + w_2 + w_3, \quad v_2 = w_2 + 3w_3, \quad v_3 = 4w_1 - 2w_2$$

(a) Find the transition matrix S from E to F .

(b) Compute the coordinate vector $[v]_F$ for $v = 2v_1 + 1v_2 - v_3$.

$$(a) \underset{\text{新基}}{[v]_F} = S \cdot \underset{\text{旧基}}{[v]_E}$$

$\rightarrow$ 旧基

① 旧基用新基表示.

② 直接拼出 S

$$v_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \rightarrow [v_1]_F \text{ (在 } F \text{ 下的坐标).}$$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} [w_1 \ w_2 \ w_3]$$

$$v_3 = \begin{bmatrix} 4 \\ -2 \\ 0 \end{bmatrix} [w_1 \ w_2 \ w_3]$$

$$S = \begin{bmatrix} -1 & 0 & 4 \\ 1 & 1 & -2 \\ 1 & 3 & 0 \end{bmatrix}$$

(b) 在新基下表示:

$$[v]_F = S [v]_E = \begin{bmatrix} -1 & 0 & 4 \\ 1 & 1 & -2 \\ 1 & 3 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -6 \\ 5 \\ 5 \end{bmatrix}$$

$$S_{E \rightarrow F} = [[v_1]_F \ [v_2]_F \ [v_3]_F]$$

把 E 的每个基向量用 F 表示, 坐标即 $[v_i]_F$.

eg. [只知道新、旧基. 不知道换算关系]

5. Consider two ordered bases $E = \{v_1, v_2, v_3\}$ and $F = \{w_1, w_2, w_3\}$ for $\mathbb{R}^3$, where

$$v_1 = \begin{bmatrix} 4 \\ 6 \\ 7 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad w_3 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

(a) Find the transition matrix S_1 from E to F .

(b) Find the transition matrix S_2 from F to E .

(c) Verify that $S_1 S_2 = S_2 S_1 = I_3$.

(d) Compute the coordinate vector $[v]_E$, where $v = \begin{bmatrix} 4 \\ 6 \\ 8 \end{bmatrix}$, and then use

S_1 or S_2 (decide by yourself) to compute $[v]_F$.

(e) Check your work by computing $[v]_F$ directly.

$$(a) \underbrace{[v_1 \ v_2 \ v_3]}_{\text{旧}} = \underbrace{[w_1 \ w_2 \ w_3]}_{\text{新}} S_1$$

$$E = FS_1, \quad S_1 = F^{-1}E. \quad \rightarrow \text{行变换.}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 4 & 0 & 0 \\ 1 & 2 & 3 & 6 & 1 & 1 \\ 1 & 2 & 4 & 7 & 1 & 2 \end{array} \right] \xrightarrow[\text{operations}]{\text{row}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & -2 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right]$$

$\hookrightarrow S_1$

$$F = ES_1$$

(b) 新旧转换

$$(c) S_1 = F^{-1}E, \quad S_2 = E^{-1}F.$$

$$S_1 S_2 = S_2 S_1 = I_3$$

(d)

$$[v_1 \ v_2 \ v_3] [v]_E = v$$

$\rightarrow v$ 可以用 E 的基与 v 在基上的坐标

在新基下表示: $E \rightarrow F$.

表示

$$[v]_F = S_1 [v]_E$$

$$(e) [w_1 \ w_2 \ w_3] [v]_F = v \quad \text{这两个 } [v]_F \text{ 肯定一样.}$$

18. 寻找4个重要子空间及其基.

eg.

6. Let $A \in \mathbb{R}^{4 \times 5}$ be

$$A = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 & c_5 \\ 1 & -2 & 1 & 1 & 2 \\ -1 & 3 & 2 & 0 & -2 \\ 0 & 1 & 3 & 1 & 4 \\ 1 & 2 & 13 & 5 & 5 \end{bmatrix}.$$

- Find the four subspaces of the matrix ($C(A)$, $C(A^T)$, $N(A)$, and $N(A^T)$, determine their bases and dimensions).
- Write down the fundamental theorem of linear algebra: Part 1 and Part 2. And verify them by the answers in (a).

Step1. 求行最简阶梯形: $\rightarrow$ 化的目标 找出列之间的换算关系..

$$A \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 0 & 7 & 3 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \triangleq R.$$

子空间1: $C(A)$: 计算列向量的关系.

$$\begin{cases} c_3 = 7c_1 + 3c_2 \\ c_4 = 3c_1 + c_2 \end{cases} \Rightarrow \{c_1, c_2, c_5\} \text{ 是列向量的基.}$$

$$\left(\begin{array}{c} A, R \text{ 关系: } C. \qquad R. \\ A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 3 & -2 \\ 0 & 1 & 4 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 7 & 3 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{(验证 } A \text{ 是3个基列向量} \\ \text{的线性组合)} \end{array} \\ \qquad \qquad \qquad c_1 \quad c_2 \quad c_5 \end{array} \right)$$

子空间2: $C(A^T)$: 取R的非零行

行变换本质上就是对行线性组合, 所以非零行反变换后必定可以得到原行向量.

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 7 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ is a basis of } C(A^T)$$

子空间3: $N(A)$: $Rx=0$

即 $Ax=0$ 令 $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$

行变换相当于方程组: $Ax=0 \Leftrightarrow Rx=0$.

$$\begin{bmatrix} 1 & 0 & 7 & 3 & 0 \\ 0 & 1 & 3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + 7x_3 + 3x_4 = 0 \\ x_2 + 3x_3 + x_4 = 0 \\ x_5 = 0. \end{cases} \Rightarrow \begin{cases} x_1 = -7x_3 - 3x_4 \\ x_2 = -3x_3 - x_4 \\ x_5 = 0. \end{cases}$$

$$So \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -7x_3 - 3x_4 \\ -3x_3 - x_4 \\ x_3 \\ x_4 \\ 0 \end{bmatrix} = \begin{bmatrix} -7 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -3 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} x_4$$

$$A \text{ basis is } \left\{ \begin{pmatrix} -7 \\ -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

子空间4: $N(A^T)$: A^T 求行变换. 之后与 $N(A)$ 同理.

$$A^T \xrightarrow[\text{operations}]{\text{row}} \begin{bmatrix} 1 & 0 & 0 & \frac{17}{4} \\ 0 & 1 & 0 & \frac{13}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 3 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + \frac{17}{4}x_4 = 0 \\ x_2 + \frac{13}{4}x_4 = 0 \\ 4x_3 + 3x_4 = 0 \end{cases} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{17}{4} \\ -\frac{13}{4} \\ -\frac{3}{4} \\ 1 \end{bmatrix} x_4.$$

$$\text{基为 } \left\{ \begin{bmatrix} -\frac{17}{4} \\ -\frac{13}{4} \\ -\frac{3}{4} \\ 1 \end{bmatrix} \right\}.$$

(b) Part 1: $\dim C(A) = \dim C(A^T) = r$

$\dim N(A^T) = m - r$. $\dim N(A) = n - r$

Part 2: $N(A) \perp C(A^T)$. $N(A^T) \perp C(A)$.

证一下.

证明正交性: 验证基——正交即可. $\rightarrow$ 结果为0

九. 证明两空间相等的套路.

eg. Show that $N(A) = S^\perp$

$\rightarrow \begin{cases} S^\perp \subset N(A) & \rightarrow \text{for any } v \in S^\perp, v \in N(A). \rightarrow \text{证明 } v \in S^\perp \text{ 条件下, } Av=0 \\ N(A) \subset S^\perp & \rightarrow \text{for any } v \in N(A), v \in S^\perp \rightarrow \text{证明 } Av=0 \text{ 条件下, } v \perp S. \end{cases}$

第三章

一. 证明 T 是线性映射 $\begin{cases} \text{可加 } T(A+B) = T(A) + T(B) \\ \text{齐次 } T(\alpha A) = \alpha T(A) \end{cases}$

二. T 的两个空间: $\text{range } T$, $\text{null } T$

$\begin{cases} \text{range } T : T \text{ 的“值域”} \\ \text{null } T : \text{使 } T(A) = 0 \text{ 的所有 } A. \end{cases}$

Rank-Nullity Theorem: $\dim(\text{range } T) + \dim(\text{null } T) = \dim(\text{输入空间})$

三. 两种映射

1. injective $\Leftrightarrow \text{null } T = \{0\}$ (仅 $A=0$ 使 $T(A)=0$).

2. surjective $\Leftrightarrow \text{range } T = V$ ($W \xrightarrow{T} V$)

四. 求线性映射代表的矩阵

eg. [求变换代表的矩阵][换基]

3. Let $E = \{u_1, u_2, u_3\}$ and $F = \{b_1, b_2\}$, where

$$u_1 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad u_3 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

and

$$b_1 = (1, -1)^T, \quad b_2 = (2, -1)^T.$$

For each of the following linear transformations L from $\mathbb{R}^3$ into $\mathbb{R}^2$, find the matrix representing L with respect to the ordered bases E and F :

$$(i) L(x) = \begin{pmatrix} x_1 + x_2 \\ x_1 - x_3 \end{pmatrix}; \quad (ii) L(x) = \begin{pmatrix} 2x_2 \\ -x_1 \end{pmatrix}.$$

Step 1: 先映射, 找出映射后的向量在 E 的坐标.

$$L(u_1) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad L(u_2) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad L(u_3) = \begin{pmatrix} 0 \\ -2 \end{pmatrix} \quad \rightarrow \text{旧基上.}$$

Step 2: 执行求转换矩阵的方法: 旧基用新基表示, 拼出 S .

$$L(u_1) = \begin{pmatrix} 1 \\ -2 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$L(u_2) = \begin{pmatrix} 3 \\ 0 \end{pmatrix} = -3 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} -3 \\ 3 \end{pmatrix}$$

$$L(u_3) = \begin{pmatrix} 0 \\ -2 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ -1 \end{pmatrix} - 2 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

$$\Rightarrow M(L, E, F) = \begin{pmatrix} 3 & -3 & 4 \\ -1 & 3 & -2 \end{pmatrix}$$

eg. [自反对基作变换]

4. Find the standard matrices A of the following linear transformations.

(a) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ is a vertical shear transformation that leaves e_2 unchanged and maps e_1 into $2e_2 + e_1$.

(b) $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ first performs a horizontal shear transformation that leaves e_1 unchanged and maps e_2 into $-2e_1 + e_2$ and then reflects points through the line $x_2 = x_1$.

$$T = (T(e_1), T(e_2))$$

$$A = \begin{pmatrix} e_1 & e_2 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$(a) T(e_1) = 2e_2 + e_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, T(e_2) = e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

(b) 两步变换: $T(A) = RH(A)$.

$$H(e_1) = e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, H(e_2) = -2e_1 + e_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad H = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$$

$$R(e_1) = e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, R(e_2) = e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\therefore T = R \cdot H = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

eg. [T代表的矩阵在不同基下的表示]

5. Let V be the subspace of $C[a, b]$ (continuous functions on $[a, b]$) spanned by $1, e^x, e^{-x}$, and let D be the differentiation operator on V .

(a) Find the matrix A representing D with respect to the ordered basis $[1, \cosh x, \sinh x]$. [$\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$.]

(b) Find the matrix B representing D with respect to $[1, e^x, e^{-x}]$.

(c) Find the invertible matrix S such that $B = S^{-1}AS$. (Hint: transition matrix)

$$[T]_{E \rightarrow F} = \begin{bmatrix} [T(u_1)]_F & [T(u_2)]_F & \dots & [T(u_n)]_F \end{bmatrix}$$

(a) 读题: 新、旧基都是 $[1, \cosh x, \sinh x]$ (不是 from... to...)

$$D(e_1) = 0 = [1, \cosh x, \sinh x] \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$D(e_2) = \sinh x = [1, \cosh x, \sinh x] \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$D(e_3) = \cosh x = [1, \cosh x, \sinh x] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

映射
旧基用新基表示

(b) 同理. 读题知新、旧基都是 $[1, e^x, e^{-x}]$

与上同理. $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

$$B = S^{-1}AS$$

五. 求相似矩阵

对线性算子 $T: V \rightarrow V$. 在旧基 E 下矩阵为 $A = [T]_E$. 在新基 F 下为 $B = [T]_F$.

$$\Rightarrow \begin{cases} B = S^{-1}AS \\ S[V]_F = [V]_E \end{cases}$$

新 旧

(这里区别于换基时转换矩阵 $[V]_F = S[V]_E$, 这里两个 S 互逆).

eg. 上例 $T_5(c)$.

S 相当于新基 $F \rightarrow$ 旧基 E 的变换矩阵.

$$F \xrightarrow{\text{对应}} B \rightarrow [1, e^x, e^{-x}]$$

(F) (E)

$$E \xrightarrow{\text{对应}} A \rightarrow [1, \cosh x, \sinh x]$$

依旧是旧基用新基表示:

$$1 = [1 \ \cosh x \ \sinh x] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$e^x = [1 \ \cosh x \ \sinh x] \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \Rightarrow S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$e^{-x} = [1 \ \cosh x \ \sinh x] \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

六. null T 条件在证明题中的常用方法.

eg.

(b) Suppose V is finite-dimensional and that $T \in \mathcal{L}(V, W)$. Prove that there exists a subspace U of V such that $U \cap \text{null } T = \{0\}$ and $T(U) = T(V)$. Find a basis.

经典方法: 对 $\forall v \in \text{null } T$, $T(v) = 0$. 可以从基的向量去证.

Let a basis of $\text{null } T$ be $v_1, \dots, v_n$. $T(v_1) = T(v_2) = \dots = T(v_n) = 0$

We can extend it to a basis of V : $v_1, \dots, v_n, u_1, \dots, u_m$.

Let $U = \text{Span}(u_1, \dots, u_m)$. Easy to know $V = U \oplus \text{null } T$

$$\Rightarrow U \cap \text{null } T = \{0\}.$$

for any $v \in U$, let $v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n + b_1 u_1 + \dots + b_m u_m$

$a_1, \dots, a_n, b_1, \dots, b_m$ are scalars.

$$T(v) = T(a_1 v_1 + \dots + a_n v_n + b_1 u_1 + \dots + b_m u_m)$$

$$= \underbrace{a_1 T(v_1) + \dots + a_n T(v_n)}_0 + \underbrace{b_1 T(u_1) + \dots + b_m T(u_m)}_{T(u)}$$

0

$T(u)$. $\forall u \in U$.

$$\Rightarrow T(V) = T(U).$$

第五章

一. 内积定义.

满足三个性质

- 非负性: $\langle x, x \rangle \geq 0$, 当且仅当 $x=0$ 取等
- 线性: 对 $y \in R^n$ 固定, $R^n \rightarrow R (x \rightarrow \langle x, y \rangle)$ 是线性的.
- 交换性: $\langle x, y \rangle = \langle y, x \rangle$

eg. Problem 1

Show that the function that takes $((x_1, x_2), (y_1, y_2)) \in R^2 \times R^2$ to $|x_1 y_1| + |x_2 y_2|$ is not an inner product on R^2 .

不线性. $B(x, y) = |x_1 y_1| + |x_2 y_2|$

Let $x = (1, 0), y = (1, 0)$.

$$B(x, y) = 1 + 0 = 1$$

$$B(-x, y) = |-1| + 0 = 1$$

$$\Rightarrow B(x, y) \neq -B(-x, y).$$

二. 内积空间

1. 性质(上面3个): 可以利用来化简内积.

$$\begin{aligned}\langle u+v, u-v \rangle &= \langle u, u \rangle + \langle u, -v \rangle + \langle v, u \rangle - \langle v, v \rangle \\ &= \langle u, u \rangle - \langle v, v \rangle\end{aligned}$$

2. 范数: $\|x\|^2 = \langle x, x \rangle$

3. 证向量相等 $\Leftrightarrow \|u-v\|=0$

三. 柯西不等式 Cauchy-Schwarz Inequality

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$$

四. 正交基.

1. 向量可以用标准正交基表示:

$$v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots + \langle v, e_m \rangle e_m$$

2. 施密特正交化 Gram-Schmidt Process

$$e_1 = \frac{v_1}{\|v_1\|}$$

$$e_j = \frac{v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}}{\|v_j - \langle v_j, e_1 \rangle e_1 - \dots - \langle v_j, e_{j-1} \rangle e_{j-1}\|}$$

3. 正交基的扩展

Problem 5

Given $x_1 = \frac{1}{2}(1, 1, 1, -1)^T$ and $x_2 = \frac{1}{6}(1, 1, 3, 5)^T$, verify that these vectors form an orthonormal set in $\mathbb{R}^4$. Extend this set to an orthonormal basis for $\mathbb{R}^4$ by finding an orthonormal basis for the null space of

$$B = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{bmatrix}$$

[Hint: First find a basis for the null space and then use the Gram-Schmidt process.]

对 x_1, x_2 的扩展, 可以把它们作为矩阵的行

$$(B = \begin{pmatrix} x_1^T \\ x_2^T \end{pmatrix}) \text{ , 再找 } \text{Null}(B) \text{ 的基}$$

(因为 $\text{Null}(B)$ 中的向量满足 $Bx = 0$, 即 x 与 x_1, x_2 均正交)

最后施密特正交化即可.

五. 正交投影

1. u 投影到 V 所在空间: $\{e_1, e_2, \dots\}$ 是目标空间的标准正交基

$$u = P_u \cdot v = \langle v, e_1 \rangle e_1 + \langle v, e_2 \rangle e_2 + \dots$$

2. 常见求正交投影的题型

① 内容相等. Problem 7

这个题中, q 是 $\cos x$
在 $P_2(\mathbb{R})$ 上的正交投影

Find a polynomial $q \in P_2(\mathbb{R})$ such that

$$\int_0^1 p(x) \overset{(v)}{(\cos \pi x)} dx = \int_0^1 p(x) \overset{(u)}{q(x)} dx$$

for every $p \in P_2(\mathbb{R})$.

② $\|v - p_U(v)\| \leq \|v - u\|$ 求最小值问题.

这个题是把 $(1, 2, 3, 4)$

投影到 U 上.

Problem 1

In $\mathbb{R}^4$, let

$$U = \text{span}((1, 1, 0, 0), (1, 1, 1, 2)).$$

Find $u \in U$ such that $\|u - (1, 2, 3, 4)\|$ is as small as possible.

把 $3x+2$ 投影到 $Ax+Bx$

上.

Problem 2

Find $p \in \mathcal{P}_3(\mathbb{R})$ such that $p(0) = 0, p'(0) = 0$, and

$$\int_0^1 |2 + 3x - p(x)|^2 dx$$

is as small as possible.

六. QR分解

设 A 的列空间有一组标准正交基 $\{e_1, e_2, \dots\}$.

$$\text{let } Q = [e_1, e_2, \dots]$$

$$A = QR. \quad (R = Q^T A)$$

(用于解决 $Ax=b$ 最小二乘问题)

$$Ax = b$$

$$QRx = b$$

$$Rx = Q^T b.$$

七. 正交补

1. 定义: $u \in \{v_1, \dots, v_m\}^\perp$

$$\Rightarrow \langle u, v_i \rangle = 0$$

2. 正交补的正交基

Problem 10

Suppose U is the subspace of $\mathbb{R}^4$ defined by

$$U = \text{span}((1, 2, 3, -4), (-5, 4, 3, 2)).$$

Find an orthonormal basis of U and an orthonormal basis of $U^\perp$.

施密特正交化得 e_1, e_2

$$\text{对 } x \in U^\perp, \langle x, (1, 2, 3, -4) \rangle = 0$$

$$\langle x, (-5, 4, 3, 2) \rangle = 0$$

$\Rightarrow$ 可以解得 $U^\perp$ 的两个基, 再施密特正交化即可.

第六章

一. 不变子空间 invariant subspace

U is invariant under T : $u \in U \Rightarrow Tu \in U$.

二. 找特征值 eigenvalue.

1. 上三角阵

Problem 6

Define $T \in \mathcal{L}(\mathbb{F}^3)$ by

$$T(z_1, z_2, z_3) = (2z_2, 0, 5z_3)$$

Find all eigenvalues and eigenvectors of T .

$$T(z_1, z_2, z_3) = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} z_1 + \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} z_2 + \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix} z_3$$

$$M(T) = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix} \Rightarrow \text{特征值 } 0, 5.$$

然后代入 $Tz = \lambda z$ 解 z .

2. 普通阵

$$\text{eg. } M(T) = A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

用 $\det(A - \lambda I) = 0$:

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{vmatrix} = \lambda^2 - 5\lambda + 4 - 6 = 0$$

3. 根据特征判断

Problem 7

Define $T: \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$ by $Tp = p'$. Find all eigenvalues and eigenvectors of T .

$$Tp = p' = \lambda p. \quad p' \text{ 与 } p \text{ 的最高项都不同, 故 } \lambda = 0.$$

$$\text{Let } Tp = \lambda p = 0 \Rightarrow p' = 0 \Rightarrow p(x) = C, \quad \underline{C \neq 0}.$$

特征向量不为0

4. 根据特征向量特征判断

Problem 9

Find all eigenvalues and eigenvectors of the backward shift operator $T \in \mathcal{L}(\mathbf{F}^\infty)$ defined by

$$T(z_1, z_2, z_3, \dots) = (z_2, z_3, \dots).$$

$$(z_2, z_3, z_4, \dots) = (\lambda z_1, \lambda z_2, \lambda z_3, \dots)$$

特征向量满足 $\Rightarrow \begin{cases} z_2 = \lambda z_1 \\ z_3 = \lambda z_2 \\ \vdots \end{cases} \quad \text{即 } z_{k+1} = \lambda z_k.$

对任意 λ , $(1, \lambda, \lambda^2, \dots)$ 是特征向量.

$$\Rightarrow c(1, \lambda, \lambda^2, \dots), \quad c \neq 0.$$

三. 用特征根法解微分方程(组)

1. 一元高阶微分方程

① 解特征根 $\lambda_1, \lambda_2, \lambda_3, \dots$

② 根为 $\begin{cases} \text{实根: } e^{\lambda_1 t}, e^{\lambda_2 t} \\ \text{二重根: } e^{\lambda t}, \lambda e^{\lambda t} \\ \text{复根 } \alpha \pm i\beta: e^{\alpha t} \cos \beta t, e^{\alpha t} \sin \beta t \end{cases}$

2. 线性微分方程组

① 一阶系统 $x' = Ax$.

I. 求 A 特征值 λ $\det(A - \lambda I) = 0$.

II. 求对应特征向量 v . $Av = \lambda v$.

IV. 每组 (λ, v) 得一个解 $x(t) = e^{\lambda t} \cdot v$

② 二阶系统 $x'' = Ax$ $e^{\lambda t}$

I. 求 A 特征值 $\lambda \rightarrow v$.

II. 设某个模式为 $x = yv$.

$$\therefore x'' = y''v:$$

$$Ax = A(yv) = y \cdot Av = y \cdot \lambda v$$

$$y''v = x'' = \lambda yv.$$

$$\Rightarrow y'' = \lambda y$$

III. 对每个特征根分别解 $y'' = \lambda y$, 再组合.

补充笔记

第二章

1. 判断向量空间 $\begin{cases} \text{满足加法, 数乘封闭性} \\ \text{满足八大公理} \end{cases}$

如 $\mathbb{R}^2$ 不是 $\mathbb{R}^5$ 的子集

2. 子空间的要求: 非空, 有 $\vec{0}$, 加法数乘封闭, 子集

3. 和空间 $U_1 + \dots + U_m$ 是 V 中包含 $U_1, \dots, U_m$ 的最小子空间

4. $U+W$ 是直和 $\Leftrightarrow U \cap W = \{\vec{0}\}$

5. S 是 $\mathbb{R}^n$ 的子空间 $\Rightarrow \dim S^\perp + \dim S = n$.

$$\mathbb{R}^n = S \oplus S^\perp$$

6. $B = S^{-1}AS$: S 是 B 的基 $\rightarrow A$ 的基的转换矩阵.
($S \in L(V)$)

第三章

1. T 是单射 $\Leftrightarrow \text{null } T = \{\vec{0}\}$

2. T 是双射 $\Leftrightarrow \dim V = \dim W$ 且 T 为单射/满射.
bijeactive

3. $\begin{cases} \dim V > \dim W \Rightarrow T \in L(V, W) \text{ 必定不是单射} \\ \dim V < \dim W \Rightarrow T \in L(V, W) \text{ 必定不是满射} \end{cases}$

4. $T \in L(\mathbb{R}^n, \mathbb{R}^m) \Rightarrow A \in \mathbb{R}^{m \times n}$
isomorphic

5. V, W 同构 $\Leftrightarrow \dim V = \dim W$.

6. 对 $T \in L(W)$: T 可逆 $\Leftrightarrow T$ 单射 $\Leftrightarrow T$ 满射

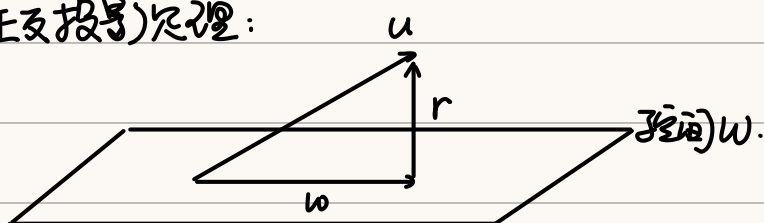
第五章

1. 0 是唯一一个与自己正交的向量

2. 柯西不等式: $|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$.

3. 平行四边形恒等式: $\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$

4. 正交投影定理:



$w = \text{proj}_W u$ 称为 u 在 W 上的正交投影.

5. 正交集: $\begin{cases} \text{只有1个向量的集合是正交集.} \\ \text{无零向量的正交集是线性无关的.} \end{cases}$

6. 正交补的性质

① U 是 V 的子集 $\Rightarrow U^\perp$ 是 V 的子空间

② $\{0\}^\perp = V, V^\perp = \{0\}$

③ $U \cap U^\perp \subseteq \{0\}$.

7. 正交矩阵 $P_W = C(C^T C)^{-1} C^T$

8. 最小二乘解: $A^T A \hat{x} = A^T b$.

或对 A 进行 QR 分解.

第二章

1. T 具有一个一维不变子空间 $\Leftrightarrow T$ 具有至少一个特征值

2. 不同特征值对应的特征向量线性无关.

3. T 可逆 $\Leftrightarrow$ 所有对角线元素均非零.

(upper-triangular matrix).

4. 对角化判断:

① $E(\lambda, T) \Rightarrow T$ 对应于 λ 的所有特征向量 + 零向量张成的空间.

② 可对角化 (T 关于 V 的某组基有对角矩阵):

$$\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_n, T)$$

5. $A \in \mathbb{R}^{m \times n} \Rightarrow A^T A$ 与 $A A^T$ 有相同的非零特征值, 它们的重数也相同.

6. 特征向量不可以是 0 向量